

Data & Knowledge Research

Research objective

- Determine what information the workbench must ingest, how that information exists in real organizations, how it can be represented, and what data is required to develop and evaluate the system.
- **What data do we actually need to prove each capability of the workbench?**

Must investigate

Enterprise data

- Engineering reports
- Inspection reports
- SOPs
- Manuals
- Internal correspondence
- Approval documents
- Financial documents
- Spreadsheets
- CSV data
- Scanned PDFs
- Engineering drawings
- P&IDs
- Photographs
- Handwritten documents
- Source code

Data characteristics

- Structured vs semi-structured vs unstructured data
- Document formats
- Document size
- Document volume
- Document quality
- OCR quality
- Image quality
- Tables
- Diagrams
- Metadata

- Document relationships
- Document versions
- Duplicate documents
- Historical documents

AI knowledge pipeline

- Document ingestion
- Document parsing
- OCR
- Layout extraction
- Chunking
- Embeddings
- Metadata
- Retrieval
- Reranking
- Source attribution
- Provenance
- Knowledge freshness

Dataset availability

Investigate:

- Public datasets suitable for the project
- Engineering-document datasets
- Technical-document datasets
- OCR datasets
- Document-understanding datasets
- Drawing/diagram datasets
- Multimodal document datasets
- Synthetic enterprise data
- Synthetic engineering documents
- Synthetic inspection reports

Keywords

R7 — Data Research

- enterprise document datasets
- engineering document datasets
- inspection report datasets
- technical report datasets
- scanned PDF datasets
- engineering drawing datasets

- P&ID datasets
- handwritten technical document datasets
- industrial document datasets
- enterprise correspondence datasets
- SOP document datasets
- engineering manual datasets
- technical drawing image datasets
- multimodal document datasets
- document OCR datasets
- document layout datasets
- table extraction datasets
- document understanding benchmarks
- visual document understanding datasets
- technical document OCR datasets
- synthetic enterprise document generation
- synthetic engineering document generation
- synthetic inspection reports
- synthetic P&ID documents
- synthetic confidential enterprise datasets
- enterprise knowledge-base data sources
- document metadata for enterprise retrieval
- document versioning for enterprise knowledge bases
- document quality assessment
- OCR quality assessment
- scanned document quality
- engineering drawing image quality
- document duplication detection
- document chunking for enterprise knowledge bases
- document indexing strategies
- document embedding datasets
- enterprise retrieval datasets
- RAG evaluation datasets
- grounded question-answering datasets
- document provenance
- source attribution in enterprise RAG
- sensitive document data handling
- confidential document anonymization
- enterprise data sanitization
- synthetic data for private AI evaluation

Output

For each dataset:

Dataset



What it contains



Format



Size



License



Relevance to project



Limitations



Possible use